controlled_root_growth_statistics

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2026-06-13

Load packages

```
library(tidyverse)
library(lme4)
library(lmerTest)
library(emmeans)
library(ggplot2)

dir.create("Controlled root growth results", showWarnings = FALSE)
```

Import data

```
df <- read.csv("controlled_root_growth_data.csv")
```

Anova

```
model2 <- aov(New.growth ~
            Species * Treatment,
          data = df)

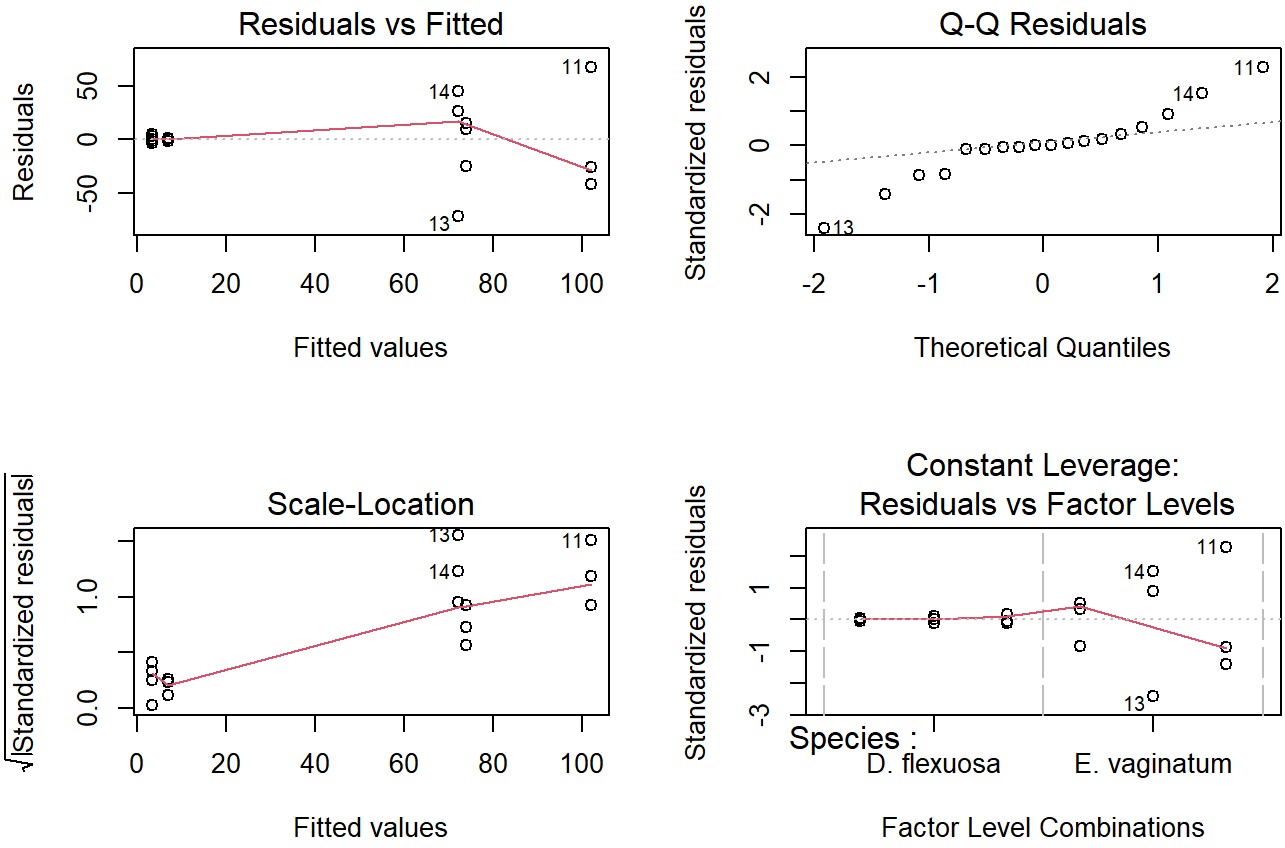
anova_results2 <- anova(model2)
anova_results2
```

```
## Analysis of Variance Table
##
## Response: New.growth
##                   Df  Sum Sq Mean Sq F value    Pr(>F)
## Species            1 27420.5 27420.5 20.4408 0.0007004 ***
## Treatment          2   757.4   378.7  0.2823 0.7589104
## Species:Treatment  2   955.5   477.8  0.3562 0.7075196
## Residuals         12 16097.5  1341.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
write.csv(
    anova_results2,
    "Controlled root growth results/ANOVA_table.csv")
```

Diagnostics

```
par(mfrow = c(2,2))
plot(model2)
```



Estimated marginal means

```
emm <- emmeans(model2, pairwise ~ Treatment | Species)

pairs_df <- as.data.frame(emm$contrasts)
pairs_df
```

```
## Species = D. flexuosa:
##  contrast                      estimate       SE df t.ratio p.value
##  Control - R chilling          3.751071 29.90492 12   0.125  0.9914
##  Control - (R+S chilling)      3.744309 29.90492 12   0.125  0.9914
##  R chilling - (R+S chilling)  -0.006762 29.90492 12   0.000  1.0000
##
## Species = E. vaginatum:
##  contrast                      estimate       SE df t.ratio p.value
##  Control - R chilling          1.650323 29.90492 12   0.055  0.9983
##  Control - (R+S chilling)    -28.164400 29.90492 12  -0.942  0.6255
##  R chilling - (R+S chilling) -29.814722 29.90492 12  -0.997  0.5926
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
write.csv(pairs_df,
          "Controlled root growth results/Species_Pairwise_Comparisons.csv",
          row.names = FALSE)
```